Azure Free Tier Setup and Virtual Machine Deployment Report

Introduction

This report documents the process of setting up an Azure Free Tier account, navigating the Azure Portal, customizing the dashboard, managing billing and costs, configuring security settings, and deploying a Virtual Machine (VM) using Microsoft Azure.

1. Azure Account Creation

To begin, I created an Azure account using the Azure for Students subscription. The registration process involved signing in with a Microsoft account, verifying my student status, and activating the Azure for Students benefits. Upon successful registration, I gained access to the Azure Portal, where cloud resources can be created and managed.

2. Azure Portal Navigation and Dashboard Customization

After logging into the Azure Portal, I explored the portal interface, which provides centralized access to all Azure services and resources. The dashboard was customized by adding useful widgets such as Resource Groups, Virtual Machines, Cost Management, and Recent Resources. This customization improved visibility and simplified resource monitoring.

3. Resource Group Creation

A Resource Group was created to logically organize and manage all resources associated with the project. Resource Groups act as containers that simplify administration, monitoring, and resource lifecycle management.

- Subscription: Azure for Students
- Resource Group: Imbaji_Cloud_Engine

4. Virtual Machine Deployment

A Virtual Machine was created using the Azure Portal.

Basic Configuration

- Subscription: Azure for Students
- Resource Group: Imbaji_Cloud_Engine
- Virtual Machine Name: Cloud Engineering
- Region: Europe (France Central)
- Availability Options: No Infrastructure Redundancy Required

• Security Type: Trusted Launch Virtual Machine
• Operating System Image: Ubuntu Server 24.04 LTS – x64 Gen2
• Architecture: x64

Administrator Account

For secure access to the virtual machine, SSH key authentication was selected.

• Authentication Type: SSH Public Key
• Username: azureuser
• SSH Key Source: Generate New Key Pair
• SSH Key Format: RSA

SSH authentication provides a more secure method of access compared to password-based authentication.

Storage Configuration

The default managed disk settings were used during deployment. Azure Managed Disks simplify storage management by automatically handling storage availability and scalability.

Networking Configuration

Azure automatically created a Virtual Network (VNet), Subnet, Network Interface, and Public IP Address. These networking components enable secure communication between the virtual machine and external clients.

Management Settings

Default management settings were retained. Azure provides capabilities such as automatic updates, backup options, and monitoring tools to support system administration.

Monitoring Configuration

Monitoring features were enabled to allow performance tracking, health monitoring, and resource utilization analysis. Azure Monitor provides insights into system performance and operational status.

Advanced Settings

Default advanced settings were used. These settings can be customized in future deployments to support specific application requirements.

Tags

Tags can be used to categorize resources based on departments, projects, environments, or cost centers. Tags improve resource organization and reporting.

Review and Create

Before deployment, Azure performed validation checks to confirm that all required configurations were correctly specified. After successful validation, the virtual machine was deployed.

5. Billing and Cost Management

To ensure responsible cloud usage, the Cost Management section of Azure Portal was accessed.

The following activities were performed:

- Reviewed Cost Analysis reports.
- Monitored current resource consumption.
- Examined projected spending.
- Explored Budget creation options.

Cost Management helps prevent unexpected charges and promotes efficient resource utilization.

6. Security Configuration

Azure follows a Shared Responsibility Model where security responsibilities are shared between Microsoft and the customer.

Microsoft Responsibilities

- Physical datacenter security
- Network infrastructure protection
- Hardware maintenance
- Platform availability

Customer Responsibilities

- Identity and access management
- Virtual machine configuration
- Data protection
- Operating system updates
- SSH key management

For this deployment, SSH key authentication was implemented to strengthen security and reduce unauthorized access risks.

7. Conclusion

The Azure Free Tier environment was successfully configured, and a Linux Virtual Machine was deployed using Azure Portal. The exercise provided practical experience in cloud resource provisioning, resource management, security implementation, cost monitoring, and Azure service administration. This deployment demonstrates fundamental cloud engineering skills and familiarity with Microsoft Azure infrastructure services.